



Progressive damage modeling in open hole composite laminates with ultrasound-informed drilling-induced delamination



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Introduction

Although drilling in composites is often required to facilitate the mechanical joining between parts, it causes delamination, burrs, and chips around the hole rim when worn-off drill bits (improper tool) or high spindle speed and low feed rates (improper setting) are selected. Drilling-induced damage may cause a premature failure of composites, and method to identify accurately the damage to predict the failure is required. Traditionally, a 2-step approach, predicting damage during drilling on a ply-by-ply basis, followed by structural simulation with applied load is employed which is computationally expensive and require validation on both steps.

Objectives: Our study aims to propose a novel framework for:

- Identifying accurately the interlaminar damage (delamination) in composite laminates due to drilling using Ultrasound Scanning system.
- Translating the identified damage into 3D finite element model as delamination in cohesive element,
- Predicting damage progression and failure onset of CFRP laminates using continuum damage mechanics.

Novel framework

Materials	NDT	FEA	Validation
16 Layer Prepreg Unidirectional CFRP laminate is prepared and drilled	In-house immersion Ultrasound system is employed to find drilling induced damage at each interface	UT informed damage is modeled for static test simulation using cohesive element and damage model	FEA simulation results are validated against experiment to affirm the integrity of proposed framework

Manufacturing, Drilling process and UT Scanning

Panels of UD carbon/epoxy T700/250F of 16-laminae with a quasi-isotropic lay-up $[0^\circ/-45^\circ/+45^\circ/90^\circ]_{2S}$ were manufactured using hot press. Samples of 250mm x 25mm were then drilled using 6.25 mm diameter drill bit (135° tip angle) at 1700 rpm to introduce delamination (Figure-1).



Figure-1. Drilling Process

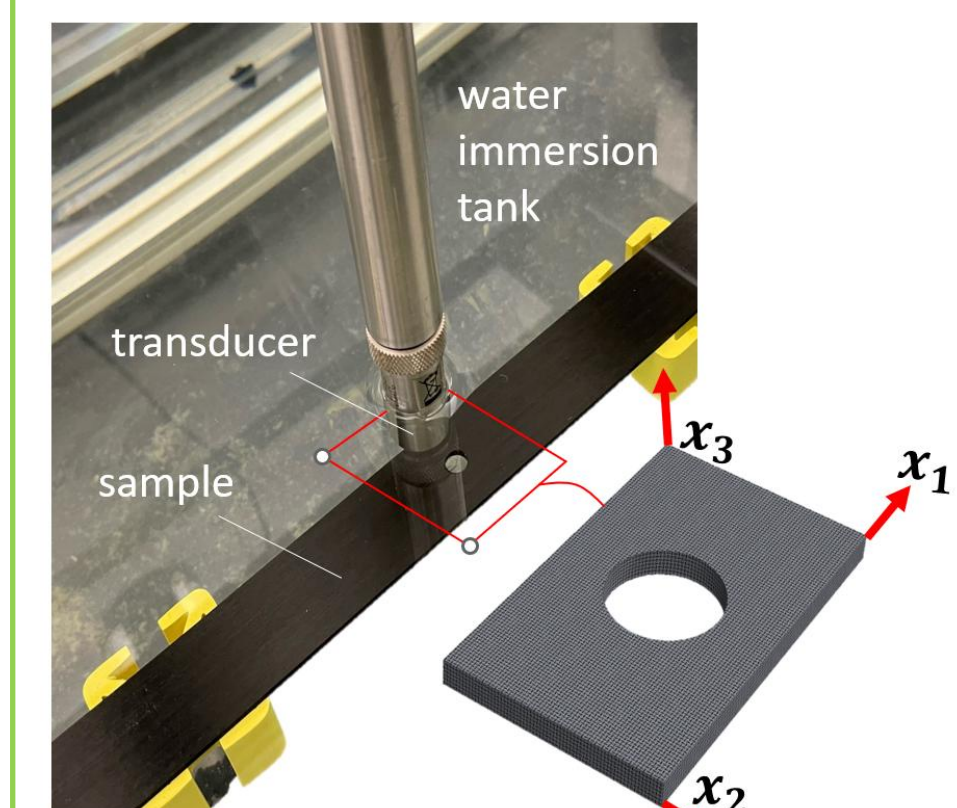


Figure-2. In-house UT immersion System

Drilling-induced delamination was scanned using in-house UT immersion system (15MHz transducer) in x_1 (scan), x_2 (index), and x_3 (focus adjustment) directions (Figure-2)

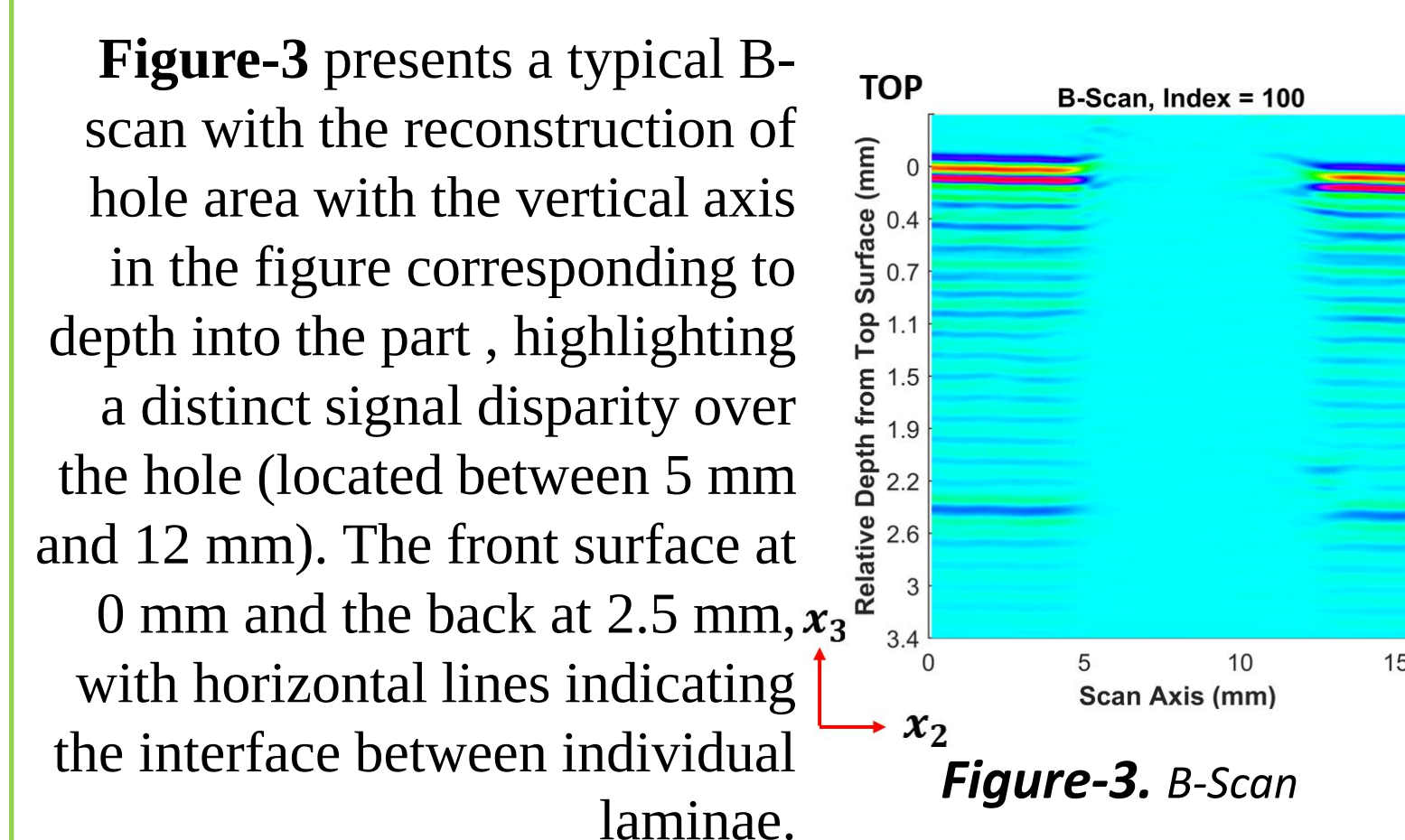


Figure-3. B-Scan

Fastener-based mechanical joining is a common practice in aerospace and automotive structures. However, in laminated composites, the drilling process for inserting fasteners can lead to non-uniform delamination. This study focuses on understanding the impact of drilling-induced delamination on the mechanical performance of carbon fiber reinforced plastic (CFRP) laminates. We utilize an in-house ultrasound testing (UT) method to quantify delamination at each ply interface. A mesoscale finite element model, developed in ABAQUS/Explicit, incorporates UT-identified delamination data for an open hole tension case study. 3D damage criteria and cohesive zone elements with a bilinear traction-separation law enable progressive damage simulation. The model accurately predicts stress concentration, damage progression, and failure in composites, aligning with experimental results. This UT-informed modeling approach offers a reliable means to assess the in-service loading capability of manufactured composites by incorporating identified damage features.

Identification of delamination using UT

At each interface, C-scan energy (Figure-4) around the hole are compared to undamaged regions. At same depth, delamination is identified by thresholding waveform's energy relative to undamaged signal resulting in a 3D profile of the damage (Figure-5).

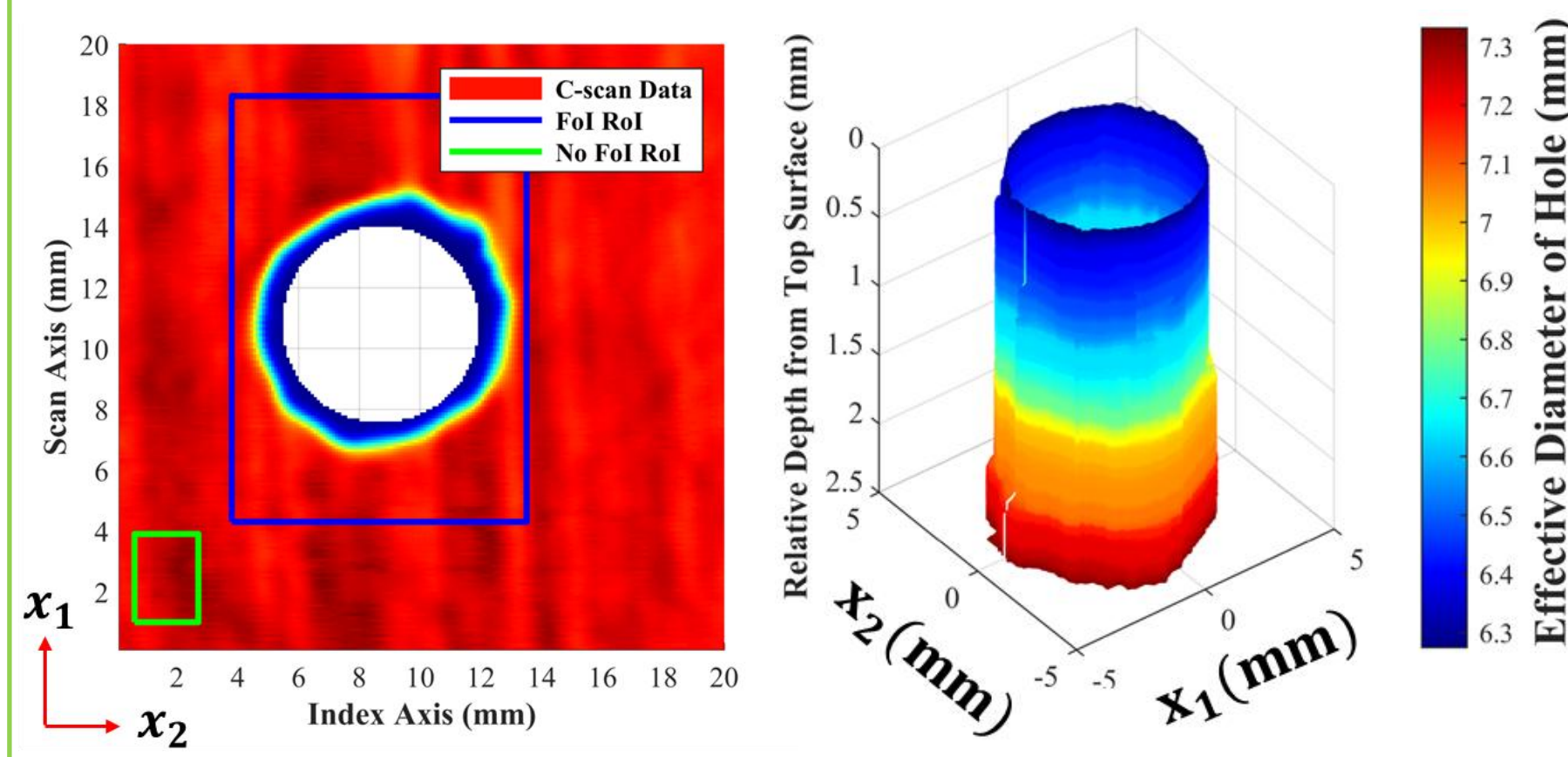


Figure-4. C-Scan

Figure-5. 3D damage area reconstruction

Effective radius (radius of the effective delaminated area) across the thickness of the 5 samples is presented in Figure-6a with damage profile for sample-1 at 3 different depths.

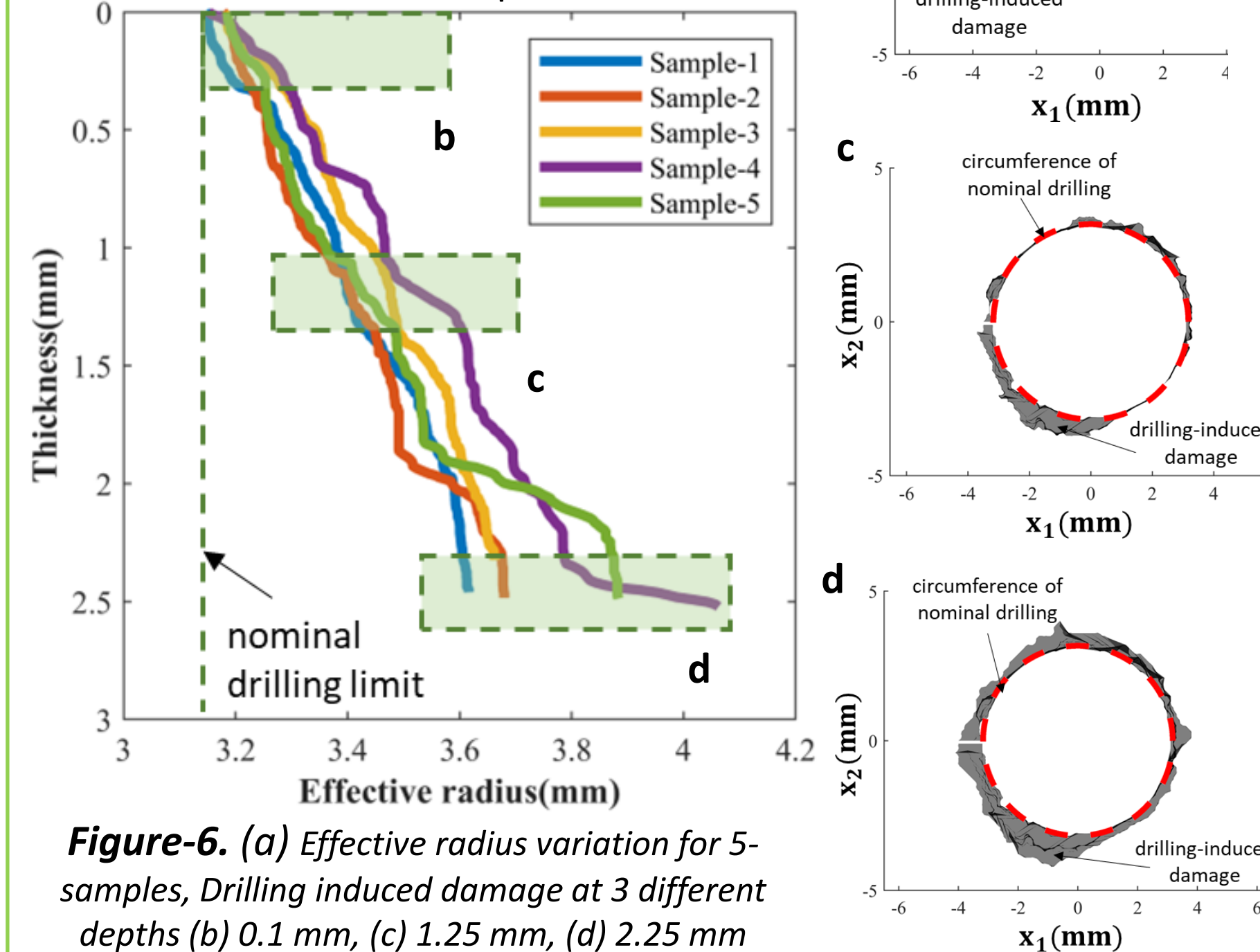


Figure-6. (a) Effective radius variation for 5-samples, Drilling induced damage at 3 different depths (b) 0.1 mm, (c) 1.25 mm, (d) 2.25 mm

Finite Element Method

- Mesoscale model (Figure-7)
- Drilling induced damage geometry information is modeled in cohesive element
- VUMAT was employed for 3D ply Failure criteria with exponential progressive damage evolution law

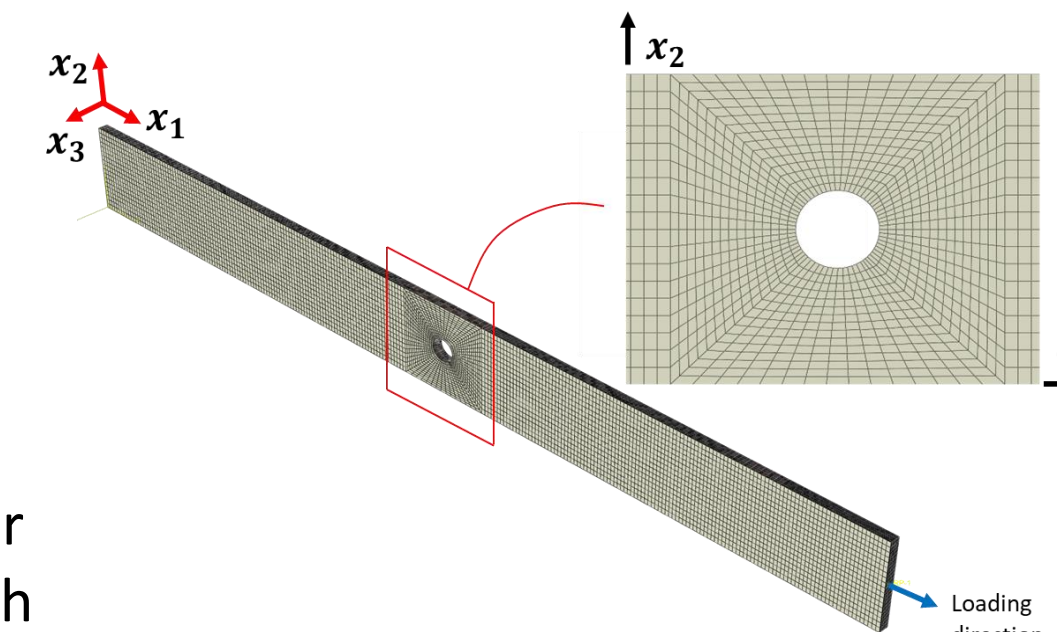


Figure-7. 3D Finite element

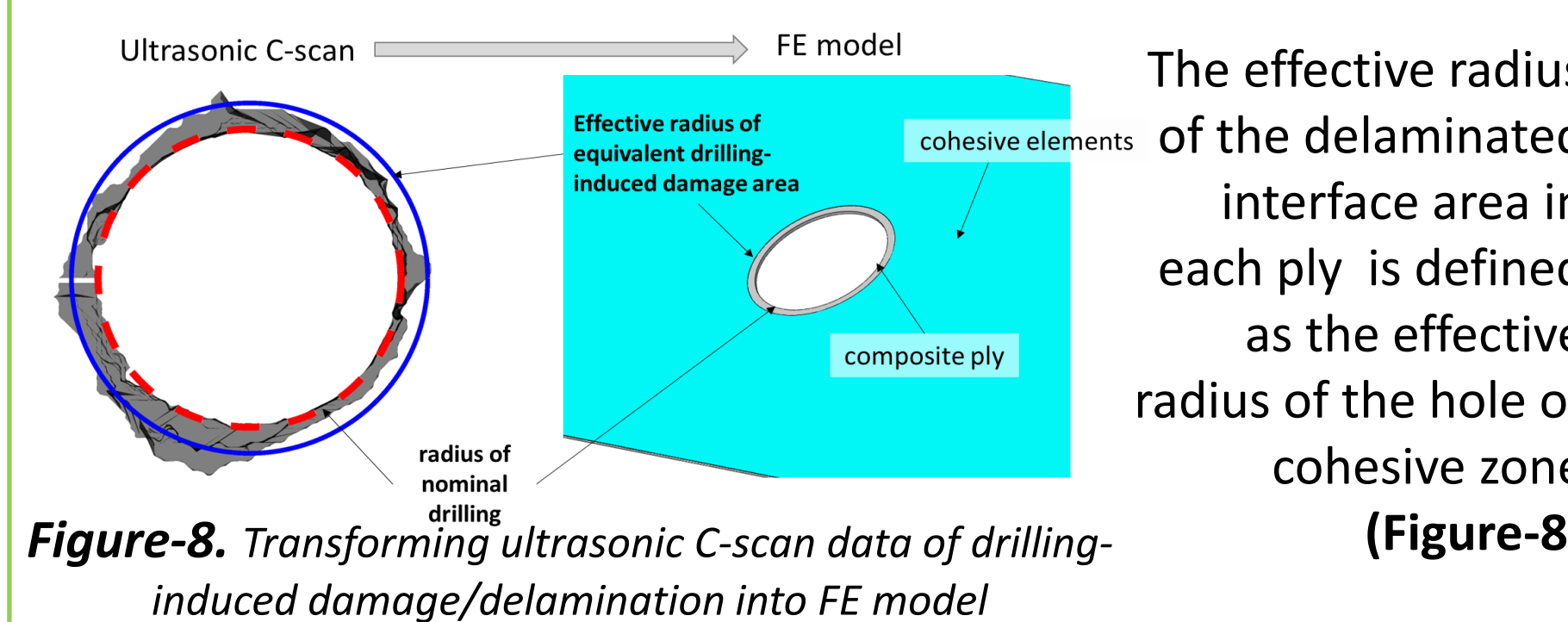


Figure-8. Transforming ultrasonic C-scan data of drilling-induced damage/delamination into FE model

The effective radius of the delaminated interface area in each ply is defined as the effective radius of the hole of cohesive zone (Figure-8)

Mechanical response of open hole tension

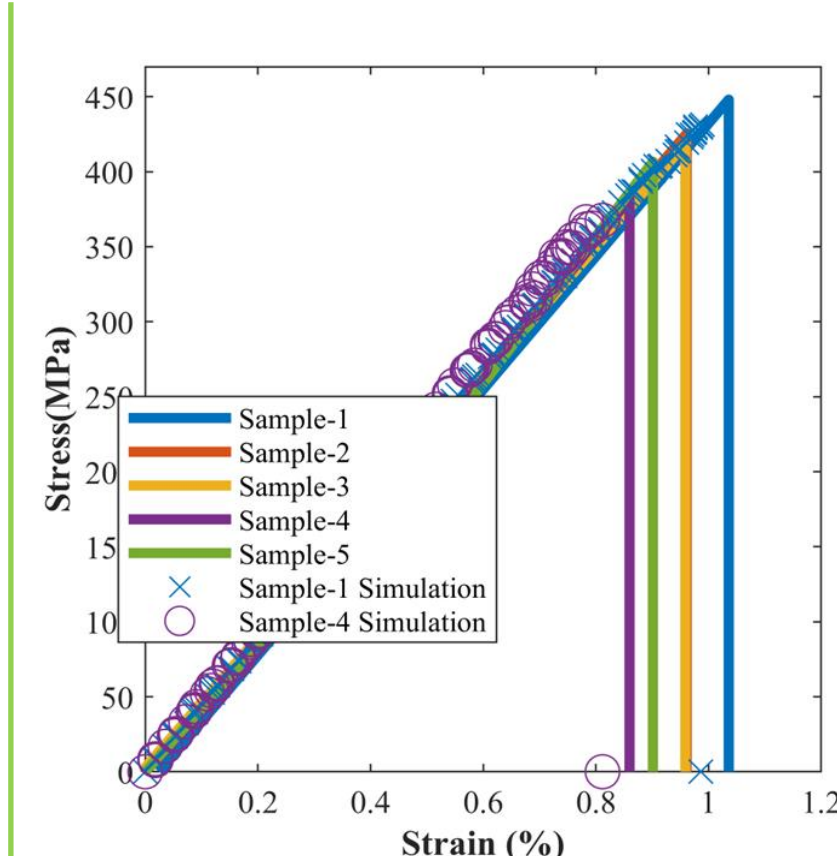


Figure-9. Stress-Strain curves of OHT samples(modelling and experiment)

All five samples exhibit a consistent stress-strain response in open-hole tensile tests (Figure-9), indicating similar effective stiffness at an average of 45.5 ± 2.3 GPa. However, there is notable variability in failure stress, averaging 416 ± 25 MPa, and failure strain at $0.94\% \pm 0.07\%$.

Correlation between effective radius and failure strain

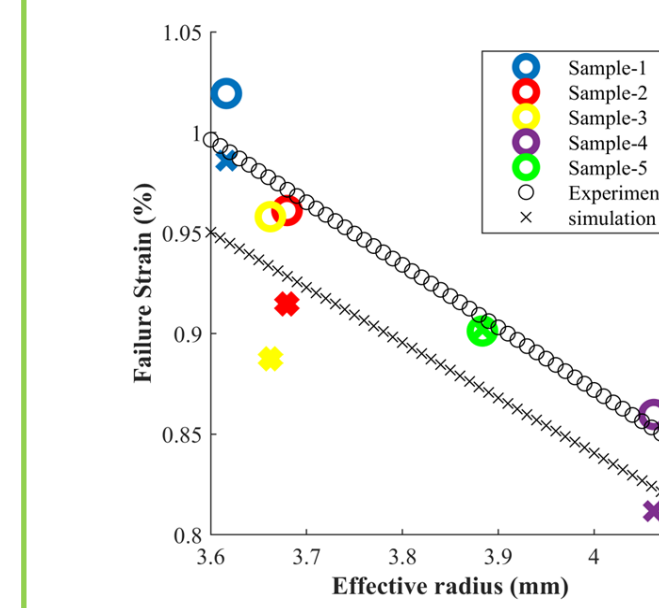


Figure-10. Correlation between failure strain and effective radius

Figure-10 shows a monotonic decrease in failure strain percentage is observed with increasing effective radius of the delamination area across all five samples in both experiment and Finite Element Method.

Strain field evolution at the hole rim

The strain field from the experimental observations using digital image correlation is compared to the observed strain field from the finite element simulations for Sample-4 shown in Figure-11.

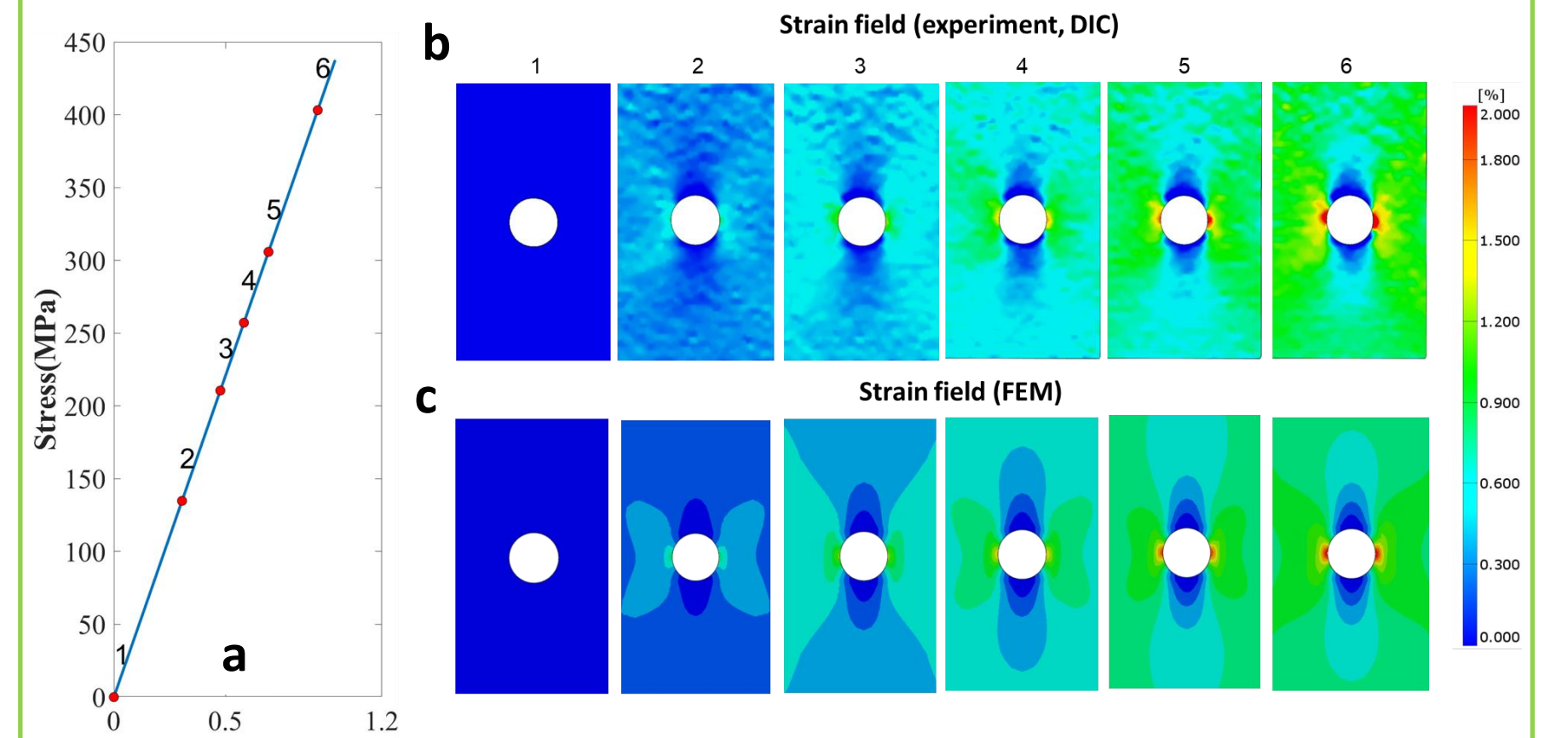


Figure-11. (a) 6 different stress level from Sample-4 OHT test, strain field evolution (b) DIC/Experiment, (c) Finite element method

The experimental and simulation stress concentration values are plotted in Figure-12 for sample-5 and the results fall between theoretical (Lekhnitskii) damaged and non-damaged stress distribution indicating the drilling damage does not fully degrade the material rather the damage is only in the interface.

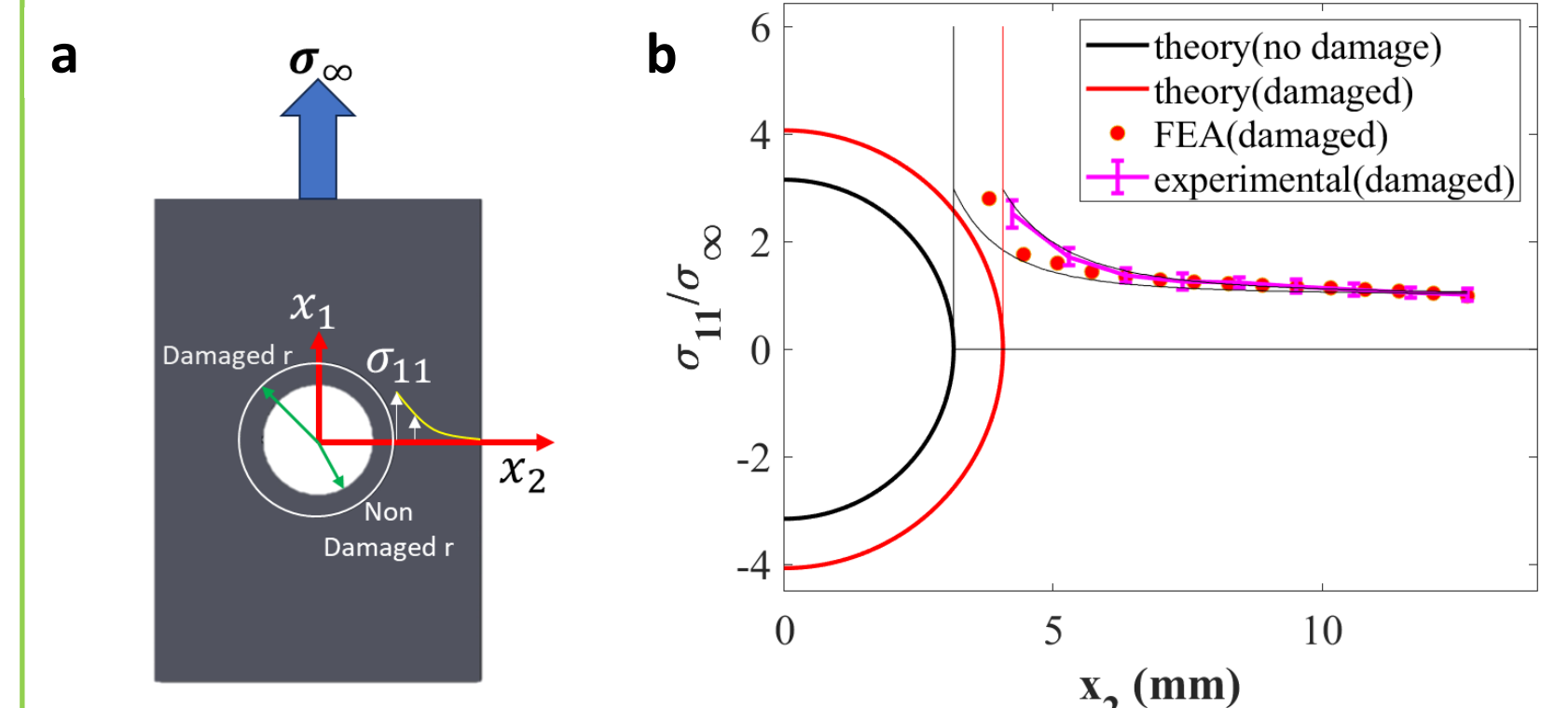


Figure-12. (a) Schematic diagram of OHT test with damaged and non damaged radii, (b) stress concentration factor from theory, experiment and FEA

Progressive damage

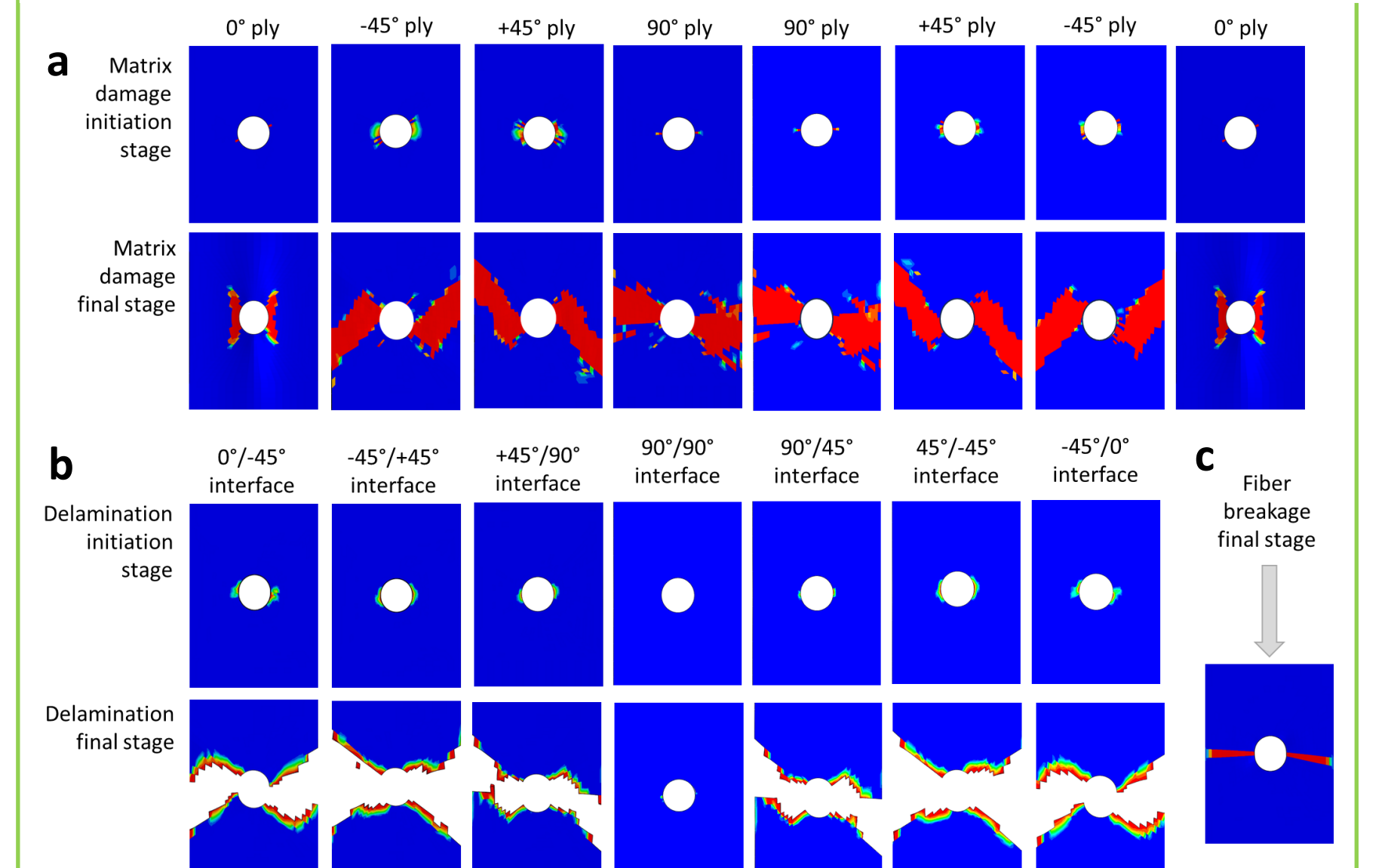


Figure-13. (a) Matrix damage, (b) Interface delamination, (c) Fiber breakage

Conclusions

We developed a novel framework for identifying interlaminar damage (delamination) in composite laminates due to drilling using UT scan, and for translating the identified damage into 3D finite element model. Cohesive element approach was used to model delamination, while ply degradation was modeled using continuum damage mechanics. The FE model was able to predict the tensile strength and corresponding damage progression during static loading. This new experimental-numerical framework will be useful for simulating similar cases involving UT-identified damage and composite failure.

Publications

Kirtunia Rahul, Arief Yudhanto, Pruthul Kekkada Ravindranath, David Jack. Progressive damage modeling in open hole composite laminates with ultrasound-informed drilling-induced delamination (Article in preparation)